

JOHNSTOWN-CAMBRIA COUNTY AP, PA, USA

WMO: 725127

Lat: 40.316N

Lon: 78.834W

Elev: 2284

StdP: 13.52

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04726

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	6.0	-6.7	4.2	4.7	-2.3	5.3	8.5	28.2	23.4	26.3	22.9	13.3	280	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	83.5	69.3	81.3	68.0	79.1	66.8	72.1	79.3	70.5	77.3	69.2	75.7	9.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	119.9	75.9	68.3	113.2	74.2	66.8	107.5	72.9	37.3	79.5	35.8	76.9	34.7	75.9	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.6	21.3	19.1	-3.2	87.7	6.4	3.4	-7.8	90.1	-11.5	92.1	-15.1	94.0	-19.8	96.5		
			-3.9	74.6	5.9	2.2	-8.2	76.1	-11.7	77.4	-15.0	78.6	-19.3	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	25.5	27.9	35.9	48.3	57.8	65.6	69.3	68.0	61.7	50.9	39.9	30.7
	DBStd	17.98	12.49	11.51	11.45	10.12	8.52	6.23	4.96	4.88	6.99	9.02	9.97	10.51
	HDD50	3055	764	622	459	150	25	0	0	0	4	101	328	602
	HDD65	6441	1224	1037	902	507	254	68	19	27	144	445	752	1062
	CDD50	2534	5	4	23	99	266	468	598	557	355	128	26	5
	CDD65	446	0	0	1	5	30	86	152	120	45	7	0	0
	CDH74	2125	0	0	2	32	165	417	751	525	215	18	0	0
	CDH80	325	0	0	0	1	15	61	136	82	29	1	0	0
Wind	WSAvg	9.2	11.7	11.4	10.8	10.3	8.5	7.8	7.1	6.5	7.1	8.9	10.1	11.1
Precipitation	PrecAvg	42.2	3.2	2.5	3.7	3.8	4.0	4.0	3.7	3.7	3.7	3.0	3.3	3.3
	PrecMax	56.7	5.6	3.9	6.2	7.8	6.3	6.1	7.0	8.5	9.3	5.7	9.0	6.7
	PrecMin	31.6	1.6	1.3	1.3	1.3	1.5	2.0	2.0	1.5	1.0	0.9	0.4	1.4
	PrecStd	6.5	1.2	0.9	1.4	1.6	1.6	1.1	1.3	1.7	2.1	1.5	1.7	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	62.6	70.6	78.7	81.9	84.8	87.3	86.0	83.6	76.6	67.4	61.1
		MCWB	53.8	52.3	56.3	59.7	65.8	69.8	71.2	70.2	67.3	64.6	55.6	53.8
	2%	DB	54.5	55.0	64.1	72.9	78.7	81.6	83.6	82.2	79.8	71.9	62.6	55.4
		MCWB	50.9	48.9	52.2	56.4	63.8	68.2	70.3	69.1	66.2	61.5	53.2	50.9
	5%	DB	50.0	49.5	58.8	68.7	75.0	79.1	81.1	79.5	76.0	68.1	58.3	51.2
		MCWB	46.4	43.7	49.8	54.8	62.2	67.3	68.5	67.9	64.8	60.0	51.0	46.9
	10%	DB	43.5	44.7	54.0	64.0	71.6	75.6	78.6	76.8	72.7	64.3	54.8	45.9
		MCWB	39.7	39.2	47.2	52.5	61.1	65.2	67.3	66.5	63.7	57.4	49.1	41.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	54.7	58.1	63.7	68.9	72.7	74.6	73.8	71.4	67.8	59.4	55.7
		MCDB	57.7	59.7	67.7	72.9	77.9	81.0	83.5	81.0	77.9	72.5	65.6	58.9
	2%	WB	51.8	50.0	54.7	59.7	66.5	70.2	72.3	71.5	69.0	64.6	55.6	51.8
		MCDB	54.2	54.1	60.5	67.4	74.4	78.3	80.0	78.0	75.2	69.6	59.8	54.9
	5%	WB	46.6	44.0	51.7	57.1	64.5	68.8	70.7	70.0	67.3	61.7	52.9	47.7
		MCDB	49.1	48.7	58.5	65.2	72.0	76.3	77.5	76.1	73.1	66.1	57.0	50.9
	10%	WB	40.1	39.4	47.0	54.5	62.5	67.4	69.3	68.4	65.5	58.4	49.5	41.8
		MCDB	43.3	43.9	53.5	62.2	69.3	74.1	75.5	74.3	70.1	63.7	54.2	45.3
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.8	15.4	18.1	16.8	15.7	15.3	15.3	15.6	14.7	13.1	11.6
		MCDBR	16.2	20.1	21.0	22.0	20.2	18.3	17.9	17.7	18.9	18.4	17.6	17.5
		MCWBR	14.2	15.6	14.1	12.2	10.3	9.2	8.0	8.1	8.8	10.3	12.3	14.6
	5% WB	MCDBR	15.8	19.2	19.6	19.4	17.4	16.0	15.4	15.3	15.0	14.9	16.2	17.1
		MCWBR	14.6	16.1	14.8	13.0	10.0	8.9	7.8	8.1	8.1	10.1	13.5	16.0
Clear-Sky Solar Irradiance	taub	0.297	0.310	0.336	0.364	0.401	0.445	0.458	0.454	0.375	0.349	0.308	0.290	
	taud	2.444	2.380	2.397	2.355	2.257	2.159	2.138	2.133	2.342	2.421	2.519	2.502	
	Ebn at Noon	276	287	291	288	278	265	260	258	274	269	270	270	
	Edh at Noon	25	31	34	38	43	47	48	47	36	30	23	22	
All-Sky Solar Radiation	RadAvg	509	774	1074	1443	1635	1796	1803	1618	1323	908	622	424	
	RadStd	44	100	95	134	154	149	116	95	155	99	72	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+95

Nomenclature: See separate page